

Meadowbrook Community Hospital

LABORATORY REPORT

Patient: Dorothy Whitfield
DOB: 11/07/1954 (Age 71)

MRN: MRN-2025-0391

Ordering Physician: Dr. Elena Vasquez, MD (Pulmonology)
Collection Date: 08/15/2026 22:40

ARTERIAL BLOOD GAS (ROOM AIR AT TRIAGE):

- pH: 7.21 CRITICAL LOW [Ref: 7.35-7.45]
- pCO₂: 88 mmHg CRITICAL HIGH [Ref: 35-45]
- pO₂: 52 mmHg (LOW) [Ref: 80-100]
- HCO₃: 34 mEq/L (HIGH) [Ref: 22-28] - chronic retention component
- Lactate: 2.4 mmol/L (HIGH) [Ref: 0.5-2.0]

CRITICAL VALUE CALLED TO ED PHYSICIAN AT 23:05

COMPLETE BLOOD COUNT (CBC):

- White Blood Cell Count: 12.8 x10³/uL (HIGH) [Ref: 4.5-11.0]
- Neutrophils: 81% (HIGH)
- Hemoglobin: 16.8 g/dL (HIGH) [Ref: 12.0-15.5] - chronic hypoxemia compensation
- Hematocrit: 50.1% (HIGH)
- Platelet Count: 301 x10³/uL (Normal)

CHEMISTRY:

- Potassium: 3.2 mEq/L (LOW) [Ref: 3.5-5.0]
- Magnesium: 1.6 mg/dL (LOW) [Ref: 1.7-2.2]
- Creatinine: 1.0 mg/dL (Normal)
- Glucose: 156 mg/dL (HIGH) - steroid effect from home prednisone burst

THEOPHYLLINE LEVEL: 6.2 mcg/mL [Therapeutic: 5-15]

CLINICAL SIGNIFICANCE:

Critical blood gas demonstrating acute-on-chronic hypercapnic respiratory failure with severe acidemia in a patient with known severe COPD. Electrolyte deficits require repletion. Findings prompted urgent BiPAP initiation and ICU evaluation.

Reported by: Critical Care Laboratory
Verified by: Dr. Michael Torres, MD, Pathologist
Date: 08/15/2026